

A quick trick for computing eigenvalues

A faster route for 2×2 matrices.

THE IDEA

For 2×2 matrices you can skip the full characteristic polynomial. Two facts do the work: the eigenvalues **sum** to $a + d$ (the diagonal, called the trace) and **multiply** to the determinant.

The trick. $m = (a + d)/2$ · $p = ad - bc$ · $\lambda = m \pm \sqrt{(m^2 - p)}$

Symbol	Meaning
m	Mean of the two eigenvalues = mean of the diagonal
p	Product of the two eigenvalues = the determinant
$\sqrt{(m^2 - p)}$	How far each eigenvalue sits from the mean

WORKED EXAMPLE

CHAPTER 15 — WORKED EXAMPLE

the fast trick for 2×2 eigenvalues

THE TRICK

$$m = \text{mean of the diagonal} = (a + d) / 2$$

$$p = \text{determinant} = ad - bc$$

$$\lambda = m \pm \sqrt{(m^2 - p)}$$

EXAMPLE

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$m = (3 + 3) / 2 = 3$$

$$p = (3)(3) - (1)(1) = 8$$

$$\lambda = 3 \pm \sqrt{(9 - 8)} = 3 \pm 1$$

$$\lambda = 4 \quad \text{and} \quad \lambda = 2$$

CHECK (always do this)

$$\text{sum of eigenvalues} = 4 + 2 = 6 = a + d \quad \checkmark$$

$$\text{product of eigenvalues} = 4 \times 2 = 8 = \det \quad \checkmark$$

Free self-check: your two eigenvalues must add up to $a + d$ and multiply to the determinant. If either fails, the answer is wrong — no answer key needed.

TEST — Chapter 15

1. A has columns (2, 1) and (1, 2). Use the trick. $\rightarrow m = 2, p = 3, \lambda = 2 \pm 1 \rightarrow \lambda = 3, 1$
2. A has columns (5, 0) and (0, 1). $\rightarrow m = 3, p = 5, \lambda = 3 \pm 2 \rightarrow \lambda = 5, 1$
3. A has columns (4, 1) and (2, 3). $\rightarrow m = 3.5, p = 10, \lambda = 3.5 \pm 1.5 \rightarrow \lambda = 5, 2$

FORMULAS FROM THIS CHAPTER

m	$(a + d) / 2$
p	$ad - bc$ (the determinant)
Eigenvalues	$\lambda = m \pm \sqrt{(m^2 - p)}$
Self-check	$\lambda_1 + \lambda_2 = a + d \cdot \lambda_1 \lambda_2 = \det$